

## How PoW, PoS, and BFT compare with each other

| Feature          | Proof of Work (PoW)         | Proof of Stake (PoS)        | BFT (Byzantine Fault Tolerance) |
|------------------|-----------------------------|-----------------------------|---------------------------------|
| Performance      | Slow transaction processing | Faster than PoW             | Very fast consensus             |
| Energy usage     | Very high                   | Low                         | Very low                        |
| Security         | Highly secure               | Secure                      | Highly secure                   |
| Scalability      | Poor scalability            | Better scalability          | Limited scalability             |
| Decentralisation | High                        | Moderate                    | Low                             |
| Cost             | Expensive                   | Lower cost                  | Low cost                        |
| Use case         | Public blockchains          | Public + modern blockchains | Private blockchains             |

users. Miners are replaced by validation users or ‘validators’ who become eligible to create a new block based on how much of the cryptocurrency is held by them in the system as their ‘stake’. Validator statuses are changed periodically.

The process includes:

- Validators lock their tokens.
  - Network selects a validator.
  - Selected validator validates transactions and creates a block.
  - Rewards are given, while malicious behaviour leads to penalties.
- PoS has the following advantages:
- Very low energy consumption compared to PoW.
  - Faster transaction validation and block creation.
  - Better scalability and higher throughput.
  - Economic penalties ensure honest behaviour.
  - Reduced need for expensive hardware.
- The disadvantages are:
- Wealth-based dominance.
  - Less proven compared to PoW in long-term scenarios.
  - Risk of centralisation among large stakeholders.
  - Initial token distribution may be unfair.

- Potential security concerns in early-stage networks.

### Byzantine Fault Tolerance (BFT)

The BFT consensus mechanism helps distributed systems work reliably and consistently, even when their nodes fail or process requests from a compromised or malicious source. To reach an agreement among the members of a distributed system, the BFT consensus mechanism requires that at least most of the participating nodes agree on the transaction to be validated. This mechanism helps to provide consistency and trust within a distributed network at the time a transaction has been committed.

The above-mentioned pros of using BFT are typical for a permissioned blockchain ecosystem in which all entities are known to each other. Because of the structured method of both communicating and validating between participants, BFT provides a high degree of reputation, rapid consensus, and a robust ability to tolerate fault in a distributed network.

The variants of BFT are:

- **PBFT (Practical Byzantine Fault**

### **Tolerance)**

- Most widely used BFT algorithm
- Works efficiently in small networks
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- **IBFT (Istanbul BFT)**
  - Improved version of PBFT
  - Used in enterprise blockchain systems
- **Tendermint BFT**
  - Combines BFT with Proof of Stake
  - Provides fast finality and high performance

## Case studies

### **Bitcoin**

- Uses Proof of Work (PoW) consensus.
- Miners solve cryptographic puzzles to validate transactions.
- Highly secure and decentralised network.
- Limitation: high energy consumption and slow processing.

### **Ethereum**

- Initially used PoW; later transitioned to Proof of Stake (PoS).
- Validators are selected based on staked tokens.